
Representations for Generation: A Survey of the Convergence of Visual Representation Learning and Generative Modeling

Kieran Didi
NVIDIA
kieran.didi@gmail.com

Abstract

Visual generation was long organized around a modular bargain: learn a perceptual compressor, freeze it, and train a generative model in its latent space. Representation learning developed in parallel, optimizing invariances for recognition rather than invertibility or likelihood. Since 2024 these lines have converged through representation-aligned denoisers, semantically shaped tokenizers, representation autoencoders, jointly learned latent priors, and native self-supervised generators. The resulting literature is difficult to compare because terms such as *representation*, *latent*, *end-to-end*, and *unified* conceal different gradient paths. We organize the field by two axes: where a representation enters the system, and how strongly representation and generation are coupled. We derive the objectives behind representative methods, isolate six properties of a useful generative latent—fidelity, semantics, information rate, modelability, geometry, and decoder robustness—and distinguish exact mathematical claims from empirical design hypotheses. We then connect the visual literature to language, multimodal systems, and molecular or 3D structure generation. Our main conclusion is deliberately non-monolithic: no single representation is universally best. The correct interface depends on which information should survive compression, which errors occur at inference, and which components are allowed to shape one another. This is a curated survey with a literature cutoff of 7 September 2026.

1 Introduction

Modern generative systems increasingly contain two learners that are easy to conflate. A *representation learner* maps data x to features z that expose useful factors of variation. A *generator* learns a probability distribution over x , or over a latent z from which x can be decoded. The former is usually rewarded for invariance; the latter ultimately pays for every visually consequential detail. Their objectives can cooperate, but they can also disagree.

Latent diffusion made this tension operational. A separately trained autoencoder first compresses images, and a diffusion or flow model then learns the latent distribution [Vahdat et al., 2021, Rombach et al., 2022]. This separation enabled efficient scaling, but it fixed the interface before the generator could express what it found easy or difficult. Meanwhile, self-supervised encoders such as DINO, DINOv2, MAE, and CLIP learned strong semantic features [Caron et al., 2021, Oquab et al., 2024, He et al., 2022, Radford et al., 2021]. The recent representations-for-generation literature asks what happens when these features supervise, initialize, replace, or co-evolve with the generative latent.

The literature is often narrated as a sequence: align denoiser features; align the autoencoder; discard the conventional VAE in favor of a foundation-model encoder; finally train representation and generation jointly. That chronology is useful, but not a taxonomy. REPA-E, UNITE, UNIFIED LATENTS (UL), and GENFIRST can all be called end-to-end while transmitting materially different gradients. SELF-FLOW learns internal features without an external vision foundation model (VFM), but does not thereby eliminate latent compression. LATENT FORCING co-denoises semantic features and pixels, yet does not learn a tokenizer. A model can share parameters without allowing every objective to update every representation.

This survey makes four contributions:

1. a two-axis taxonomy based on *intervention point* and *training coupling*;
2. a common mathematical language for auxiliary alignment, pretrained-feature latents, rate-controlled joint latent learning, and native self-supervision;
3. an evaluation framework that separates six latent properties and explicitly traces gradient flow; and
4. cross-domain case studies showing what changes in language, unified multimodal models, and molecular or 3D structure generation.

Scope and evidence. We focus on diffusion, flow-matching, and autoregressive systems for which learned representations are a primary design variable. This is a curated technical survey, not a systematic review or a leaderboard. Many 2026 results are preprints with incomparable compute, guidance, sampling, and evaluation protocols. We label a company announcement as product evidence rather than as a reproducible method, and use social-media posts only for discovery and context. Unless otherwise stated, numerical claims belong to the cited paper’s own protocol.

Central thesis. The choice of latent is a constrained interface-design problem. A useful code must preserve what the decoder cannot cheaply hallucinate while presenting a distribution that the prior can learn. Reconstruction and recognition probe only parts of that contract. Information rate, covariance and spectral structure, manifold geometry, exposure to off-manifold inputs, and the exact gradient path can dominate.

2 A Common Mathematical Language

2.1 The modular latent generator

Let data $x \sim p_{\text{data}}$, encoder $q_\phi(z \mid x)$, decoder $p_\psi(x \mid z)$, and learned prior $p_\theta(z)$. A modular latent generator performs

$$x \xrightarrow{q_\phi} z \quad \text{and} \quad z_0 \sim p_\theta \xrightarrow{p_\psi} \hat{x}.$$

The interface z serves two clients with different wishes. The decoder prefers a code rich enough for faithful reconstruction. The prior prefers a regular, low-complexity distribution. A semantic learner may prefer invariance to texture, pose, or instance detail. Latent quality therefore cannot be a single scalar.

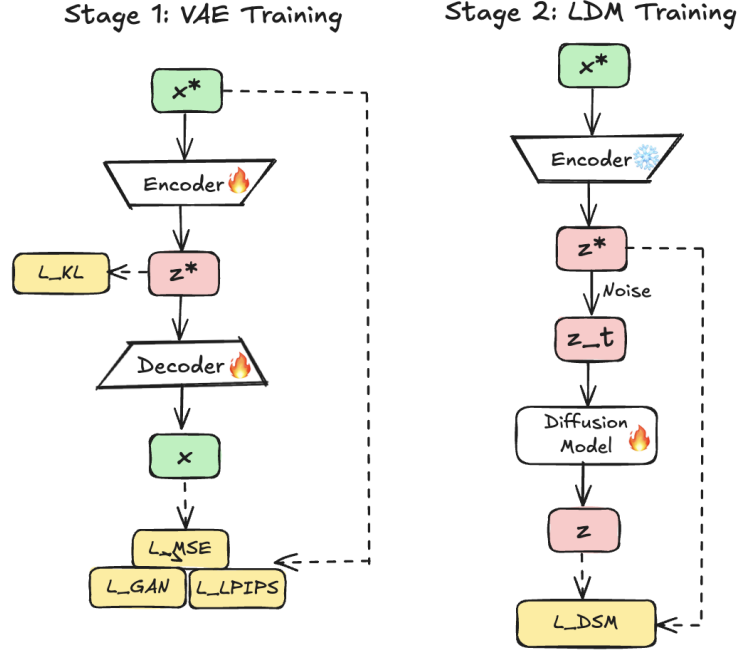


Figure 1: The modular latent-diffusion recipe: train the autoencoder and latent prior in separate stages. Author-created Excalidraw diagram based on the latent diffusion architecture [Rombach et al., 2022].

For a variational autoencoder (VAE), introduce an auxiliary posterior $q_\phi(z | x)$. The marginal log likelihood obeys

$$\begin{aligned} \log p_{\theta, \psi}(x) &= \log \int q_\phi(z | x) \frac{p_\theta(z) p_\psi(x | z)}{q_\phi(z | x)} dz \\ &\geq \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\psi(x | z)]}_{\text{reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) || p_\theta(z))}_{\text{rate / prior match}} \equiv \mathcal{L}_{\text{ELBO}}(x), \end{aligned} \quad (1)$$

where the inequality is Jensen’s. The color code will be used throughout: **blue** denotes encoder or representation terms, **orange** denotes prior or generative terms, **green** denotes decoder terms, and **purple** denotes external or target representations.

Equation (1) exposes the classic trade-off but hides two practical facts. First, perceptual and adversarial reconstruction losses need not correspond to a normalized likelihood. Second, a simple analytic prior such as $\mathcal{N}(0, I)$ can be replaced by a learned diffusion prior. The resulting system can be trained jointly, in stages, or with selected stop-gradients. These choices change the optimization problem even when the forward graph looks identical.

2.2 Diffusion and flow objectives

In a variance-preserving diffusion parameterization [Ho et al., 2020, Song et al., 2021],

$$z_t = \alpha_t z_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad \alpha_t^2 + \sigma_t^2 = 1. \quad (2)$$

A denoiser may predict noise, clean data, score, or velocity. For noise prediction,

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{z_0, t, \epsilon} \left[w(t) \|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2 \right], \quad (3)$$

with optional condition c . Different $w(t)$ correspond to different variational or perceptual emphases. The apparent simplicity of Equation (3) is deceptive: rescaling z_0 , changing its covariance, or

concentrating it near a manifold changes the effective signal-to-noise ratio and the geometry traversed by the probability path.

Flow matching uses a conditional interpolation, commonly [Lipman et al., 2023]

$$z_t = (1 - t)z_{\text{noise}} + tz_{\text{data}}, \quad u_t = z_{\text{data}} - z_{\text{noise}},$$

and minimizes

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} [\|v_\theta(z_t, t, c) - u_t\|_2^2]. \quad (4)$$

The learned vector field defines the ODE $dz_t/dt = v_\theta(z_t, t, c)$. Straight Euclidean paths are convenient, not neutral: if codes occupy a curved or anisotropic set, interpolants may leave high-density regions.

2.3 Representation alignment

Let $h_\theta^\ell(z_t, t)$ be a hidden denoiser feature and $r_\omega(x)$ a frozen teacher feature. A representative alignment objective is

$$\mathcal{L}_{\text{align}} = \mathbb{E} \left[1 - \frac{g(h_\theta^\ell)^\top \text{sg}(r_\omega(x))}{\|g(h_\theta^\ell)\|_2 \|\text{sg}(r_\omega(x))\|_2} \right], \quad \mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda(t, k)\mathcal{L}_{\text{align}}. \quad (5)$$

The projector g , layer ℓ , teacher, spatial correspondence, and schedule λ are all substantive design choices. Because the teacher is stopped, alignment changes the denoiser, not the target representation. If λ is switched off early, as in HASTE, the term is better understood as an optimization curriculum than as a permanent semantic constraint [Yu et al., 2024, Wang et al., 2025].

2.4 Trace the gradients, not the slogans

For a joint objective

$$\mathcal{J}(\phi, \theta, \psi) = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{prior}}\mathcal{L}_{\text{prior}} + \lambda_{\text{rep}}\mathcal{L}_{\text{rep}}, \quad (6)$$

the crucial question is whether

$\nabla_\phi \mathcal{L}_{\text{prior}}$ is active, stopped, approximated, delayed, or mediated only by shared weights.

If it is active, the generator can reshape the encoder. That can improve modelability, but it can also collapse the code to make prior fitting easy. If it is stopped, representation and generation may still interact through a shared trunk or through alternating tasks. Joint training should therefore be reported as a gradient-routing matrix, not a Boolean flag.

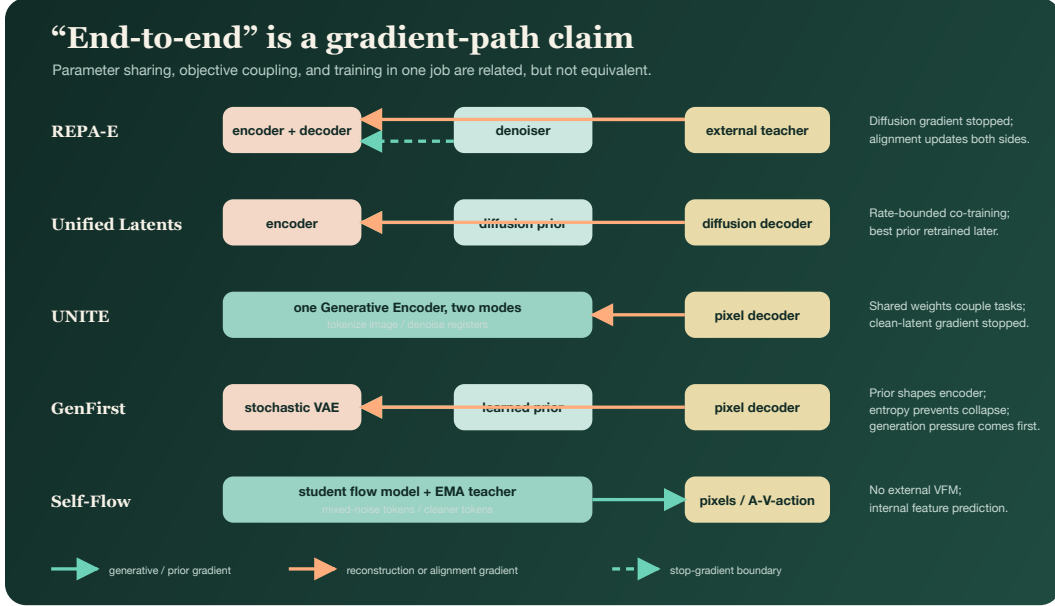


Figure 2: Original comparison of dominant gradient paths. Similar forward pipelines can implement different optimization problems. A stop-gradient on the clean code does not imply complete independence when parameters are shared elsewhere.

3 From Parallel Literatures to a Two-Axis Taxonomy

3.1 Historical compression

The VAE formalized amortized latent-variable learning [Kingma and Welling, 2014]; VQ-VAE supplied a learned discrete tokenizer [van den Oord et al., 2017]. Score-based latent models showed that a learned diffusion prior could be integrated with a hierarchical autoencoder [Vahdat et al., 2021]. Latent diffusion then established the modular engineering recipe: train a perceptual compressor, freeze it, and learn a denoiser in the compressed space [Rombach et al., 2022]. DiT made the latent denoiser a scalable Transformer [Peebles and Xie, 2023].

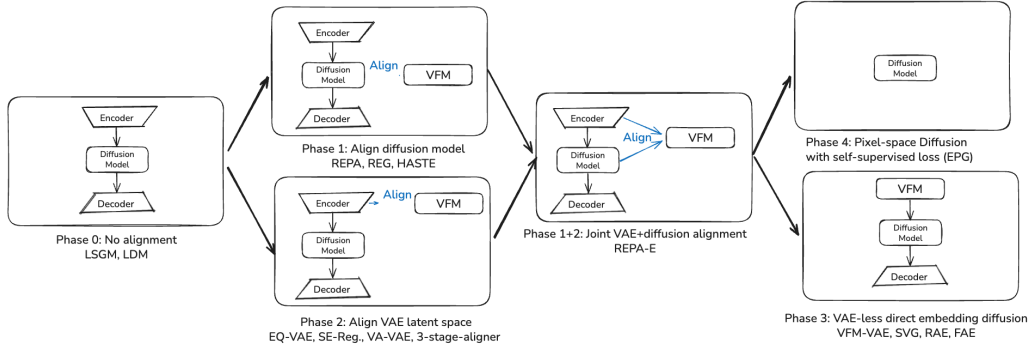
In parallel, self-supervised visual encoders developed increasingly transferable semantic spaces. Their features were usually evaluated by linear probes, retrieval, segmentation, or robustness, not by whether pixels could be reconstructed or whether a generative prior could model their full joint distribution. The convergence began when generative models used those features as internal targets. REPA aligned DiT hidden states to frozen semantic features and reported substantially faster optimization [Yu et al., 2024]. Follow-up work varied targets, schedules, and coupling, while another branch moved the alignment into the tokenizer. A third branch replaced the conventional VAE encoder with a VFM. The 2026 wave made the final step explicit: allow representation, prior, and decoder to shape one another.

3.2 The taxonomy

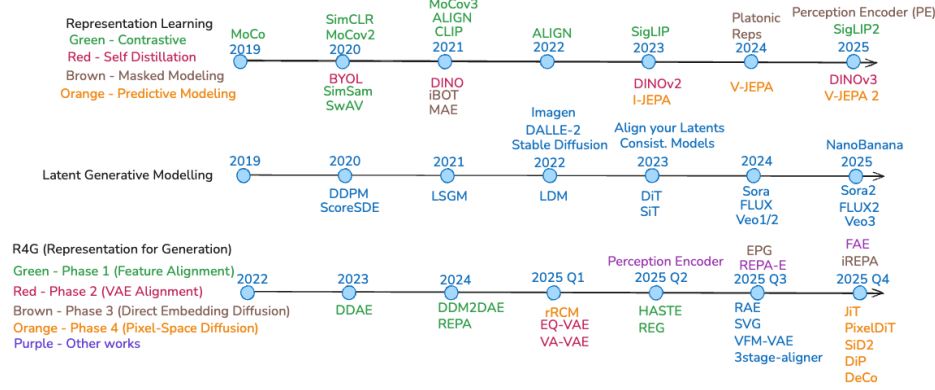
Figure 4 separates two questions.

Where does the representation enter? (i) as an auxiliary target inside a denoiser; (ii) as a constraint on a tokenizer or VAE; (iii) as the generative latent itself; or (iv) inside a native pixel/data-space or multimodal model. These positions imply different inference costs and different failure modes.

How strongly is training coupled? A representation can be frozen, selectively adapted or staged, or jointly learned. Parameter sharing, gradient sharing, and schedule coupling are distinct. A frozen teacher can still strongly shape early optimization; a shared encoder can receive stopped clean-code gradients; a joint first stage can still be followed by a freshly trained prior.



(a) The original post’s four-phase storyline: from unaligned latent generation to joint/native representation learning.



(b) A chronological snapshot of representation learning, latent generation, and representations-for-generation through the first RAE wave.

Figure 3: Two complementary views retained from the accompanying article. The phase view is a teaching narrative; the timeline records precedent. Neither replaces the mechanism-first taxonomy in Figure 4. Author-created Excalidraw diagrams.

Family	Representation enters as	Typical coupling	Inference consequence	Diagnostic question
Denoiser alignment	auxiliary hidden-state target	frozen teacher; scheduled weight	usually none	Does alignment accelerate useful features or overconstrain them?
Aligned Representation AE	regularizer or target for encoder pretraining feature is the code	frozen teacher; encoder adapted frozen/lightly adapted encoder	same pipeline decoder must invert VFM space	Which semantic invariances can reconstruction tolerate? Are semantic codes complete, normalized, and locally decodable?
Joint latent learning	learned interface among encoder, prior, decoder	shared or routed gradients	architecture dependent	Can prior pressure improve modelability without collapse?
Native/self-supervised	target generated by model or trajectory	joint/EMA	may remove external teacher	What asymmetry prevents a trivial target?

Table 1: Mechanism-first comparison. The final column is more portable than any method label.

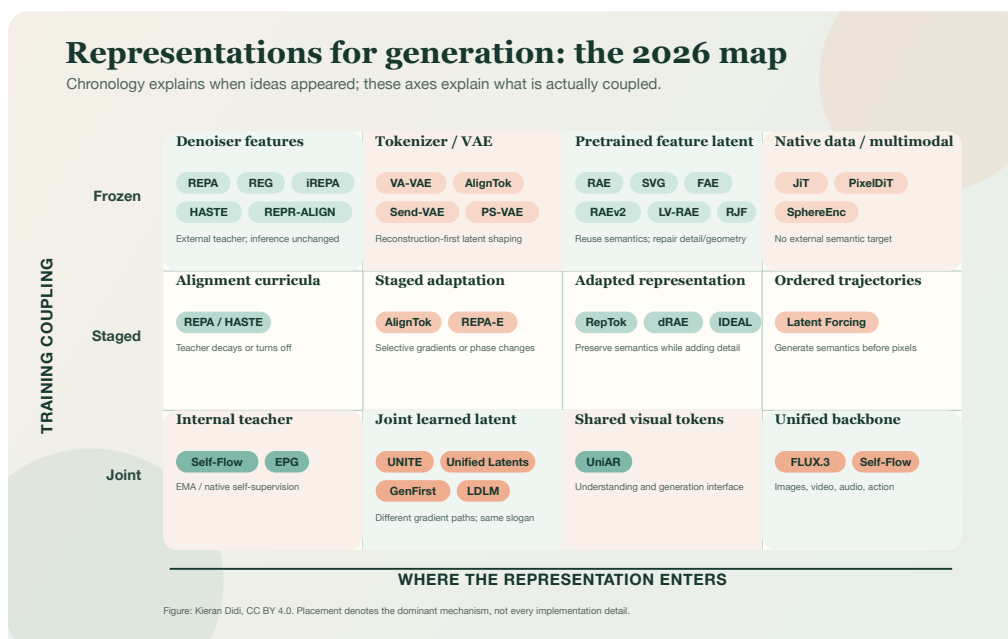


Figure 4: A two-axis map of representative methods. Placement denotes the dominant mechanism rather than every implementation detail. Chronology explains when ideas appeared; these axes explain what is coupled. Original figure, CC BY 4.0.

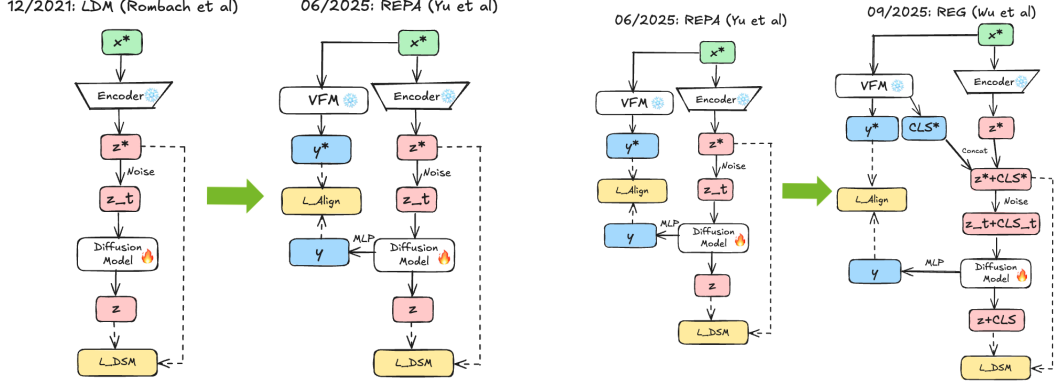
4 Auxiliary Representations Inside the Generator

REPA adds a semantic prediction task to intermediate denoiser features while leaving the VAE and inference graph unchanged [Yu et al., 2024]. This is attractive because the auxiliary loss is removable: one can improve optimization without paying for the teacher at sampling time. It also creates a clean experimental question about what feature target helps.

The answer is not simply the strongest recognition model. WHAT MATTERS FOR REPRESENTATION ALIGNMENT finds that spatial self-similarity is highly predictive in its controlled comparison, while global semantic quality alone is insufficient [Singh et al., 2025]. The result is specific to using a representation as a patchwise denoiser target. It does not imply that global semantics are unimportant when the representation is the latent itself. Indeed, RAEv2 reports benefits from globally semantic encoders while an added REPA-style objective improves spatial organization [Singh et al., 2026]. The location of the representation changes the desired property.

Permanent alignment can also become a liability. HASTE reports that turning alignment off early preserves its optimization benefit while avoiding late-stage conflict [Wang et al., 2025]. REG entangles representation guidance more deeply with generative features [Wu et al., 2025]. These variants suggest three interpretations of Equation (5): representation alignment can be (i) a semantic regularizer, (ii) a preconditioner or curriculum that shapes early features, or (iii) an internal guidance signal. Experiments should distinguish them by sweeping the active interval, target layer, spatial granularity, and teacher.

Practical recipe. Normalize target and predicted features, isolate the alignment head from the output head, and report whether the teacher operates on clean images, decoded latents, or augmented views. Log the cosine loss by diffusion time. If alignment helps only at high noise or only early in training, a scheduled objective is more faithful than an always-on penalty. Evaluate generation after removing the head to verify that the claimed gain is carried by the denoiser rather than by an inference-time crutch.



(a) REPA adds a removable VFM-alignment target to the denoiser.

(b) REG feeds representation-derived information more deeply into generation.

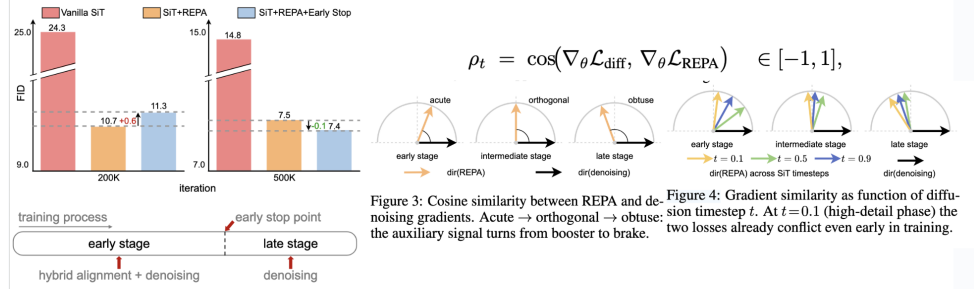


Figure 3: Cosine similarity between REPA and denoising gradients. Acute $\rightarrow$ orthogonal $\rightarrow$ obtuse: the auxiliary signal turns from booster to brake.

(c) HASTE turns alignment off after an early phase, separating feature acquisition from late denoising refinement.

Figure 5: The Phase-1 alignment lineage: REPA [Yu et al., 2024], REG [Wu et al., 2025], and HASTE [Wang et al., 2025]. The forward graph changes less than the duration and role of the representation signal. Author-created Excalidraw diagrams; the HASTE panel includes small adapted result panels from the cited work, which are excluded from this paper’s CC BY license.

5 Shaping the Tokenizer and Adopting Pretrained Features

5.1 Semantically aligned autoencoders

Conventional perceptual autoencoders already borrow representation losses through LPIPS-like feature distances, but recent methods place semantic organization at the center. REPA-E allows a representation objective to update the VAE encoder while stopping the diffusion loss from flowing into that encoder [Leng et al., 2025]. Thus it is end-to-end in scheduling and system assembly, not in the strongest gradient-sharing sense. ALIGNTOK adapts a VFM toward a compact tokenizer while trying to preserve useful semantic structure [Chen et al., 2025]. PS-VAE argues that both semantics and reconstruction are needed for text-to-image generation and editing [Zhang et al., 2025]. Work on autoencoder diffusability likewise shows that reconstruction FID can improve while the latent distribution becomes harder to model [Skorokhodov et al., 2025].

VA-VAE makes the reconstruction-generation tension explicit by aligning the learned VAE code to a frozen vision foundation model while retaining pixel, perceptual, adversarial, and KL objectives [Yao et al., 2025]. This is still a learned compressor: the VFM is a target, not the inference-time encoder. A related staged family starts from VFM features and progressively adapts an adapter and decoder. ALIGNTOK’s three stages—latent alignment, perceptual alignment, and decoder refinement—separate semantic preservation from fine-detail recovery rather than asking one joint loss to solve both at once [Chen et al., 2025].

The optimization conflict is easy to state. Suppose a nuisance variable n affects pixels but not category. A representation objective may reward

$$I(z; y) \text{ large,} \quad I(z; n) \text{ small,}$$

whereas an invertible decoder needs enough information about n to reconstruct x . A practical tokenizer can allocate low-frequency or semantic information to an aligned subspace and residual detail to another, but this is an architectural compromise rather than a proof that semantics and fidelity are naturally identical.

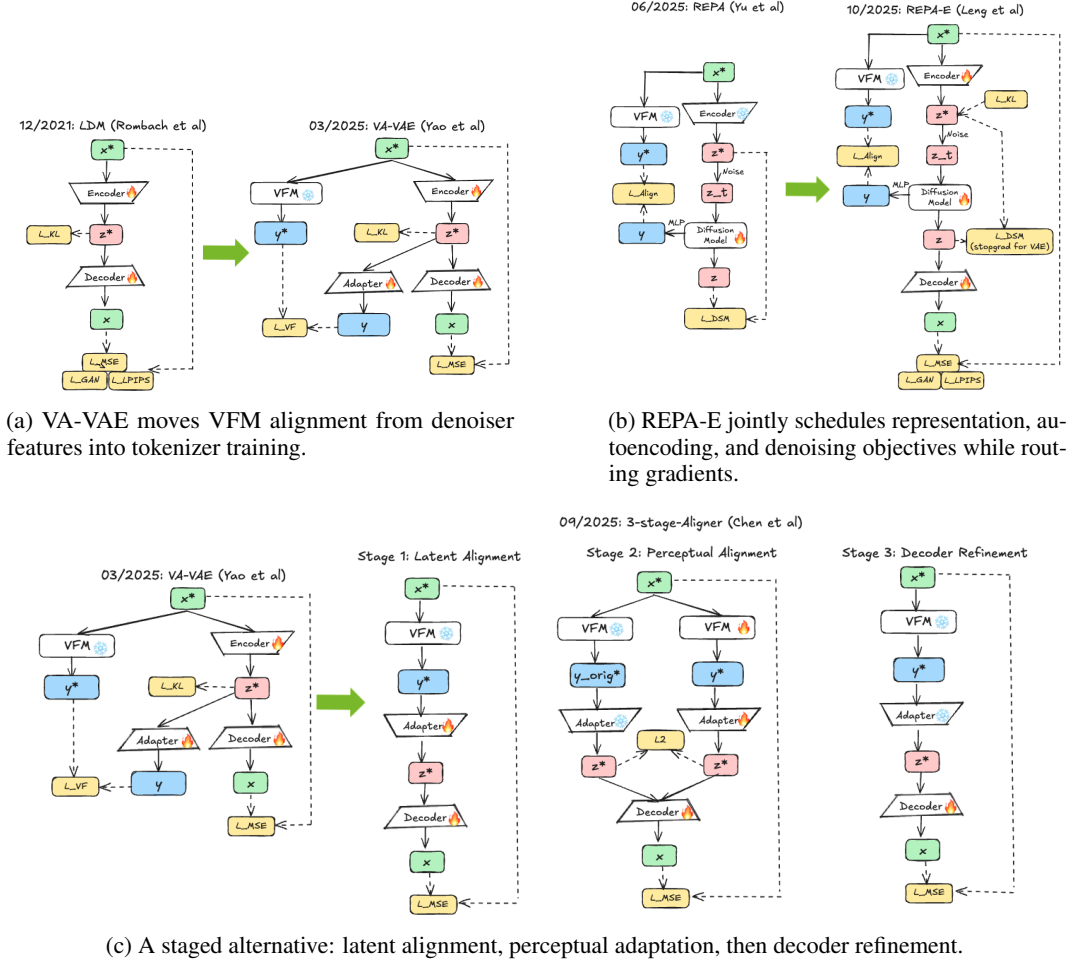


Figure 6: Ways to shape the tokenizer with representations: VA-VAE [Yao et al., 2025], REPA-E [Leng et al., 2025], and the three-stage AlignTok recipe [Chen et al., 2025]. These systems can look similar at inference while optimizing different gradient graphs. Author-created Excalidraw diagrams.

5.2 Representation autoencoders

A representation autoencoder (RAE) uses a pretrained encoder E_{VFM} as the image code and trains a decoder D_ψ to reconstruct pixels. The prior learns

$$z_0 = E_{\text{VFM}}(x), \quad \mathcal{L}_{\text{prior}} = \mathbb{E} \|v_\theta(z_t, t) - u_t\|^2.$$

The promise is compelling: replace a low-level VAE code with a semantically organized representation and inherit the scaling of foundation-model pretraining. The cost is that the code may be high-dimensional, anisotropic, lossy in local detail, or concentrated near a curved manifold. A decoder trained only on exact encoder outputs may also be brittle to the approximate samples produced by the prior.

The original RAE work couples a frozen pretrained encoder with a learned pixel decoder and adapts the generative architecture to the resulting wide latent [Zheng et al., 2025]. Scaling studies show that conclusions depend on DiT width and encoder dimension [Tong et al., 2026]. RAEv2 aggregates several encoder layers, compares 27 encoders, and combines the latent with internal representation

guidance; its main lesson is that RAE and REPA are complementary, not mutually exclusive [Singh et al., 2026]. LV-RAE restores low-level information and explicitly trains decoder robustness around the feature manifold [Liu et al., 2026].

Two intermediate designs explain why “replace the VAE” is not a single operation. VFM-VAE merges several frozen VFM layers, compresses them through a regularized latent, and reconstructs progressively at multiple scales [Bi et al., 2025]. SVG instead combines a frozen semantic feature with a learned residual branch for details that the VFM discarded [Shi et al., 2025]. RAE removes both the additional learned semantic alignment and the mandatory residual encoder: a frozen pretrained representation is the code, and the burden shifts to the decoder and prior.

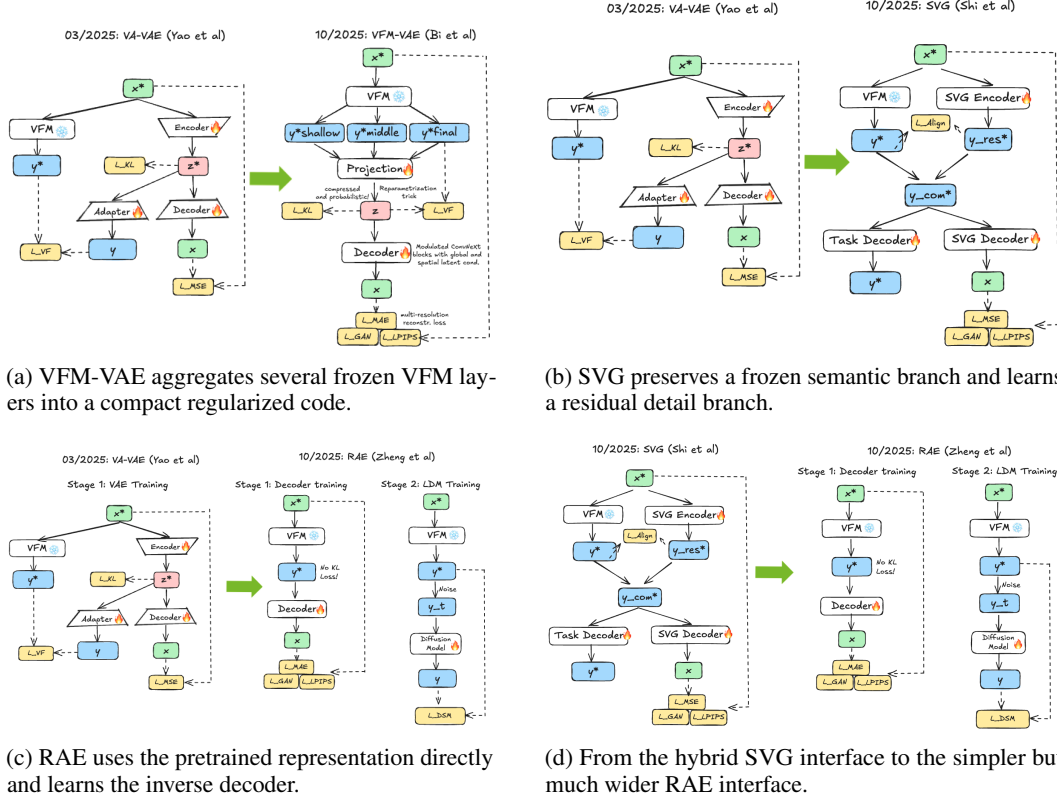


Figure 7: The Phase-3 design progression from aligned learned codes to pretrained-feature latents [Yao et al., 2025, Bi et al., 2025, Shi et al., 2025, Zheng et al., 2025]. These are intervention-point comparisons, not claims that every later method strictly subsumes every earlier one. Author-created Excalidraw diagrams.

FAE is a useful boundary case. It keeps the VFM frozen but compresses its patch features using one self-attention layer plus a projection. A six-layer feature decoder reconstructs the original VFM feature field using feature MSE and a small KL term; a separate pixel decoder is trained on noisy original features and then refined on reconstructed features [Gao et al., 2026b]. Only after this staged adaptation is a generative prior learned. It is therefore neither one projection and done nor a fully joint generator. Its claim is that a very small trainable bottleneck can preserve useful VFM geometry while restoring a conventional, compact generative interface.

Discrete and spherical variants. IDEAL aligns discrete codes with shallow and deep VFM features [Chen et al., 2026]; DRAE uses hyperspherical codes to reduce sensitivity to feature magnitude [Ma et al., 2026]. RAE-AR addresses sequential generation by normalizing codes and injecting Gaussian noise during autoregressive training [Yu et al., 2026]. Here exposure bias has its sequence-model meaning: training sees ground-truth previous codes, whereas generation conditions on its own imperfect samples. This differs from poor reconstruction on clean features and from geometric path mismatch, even though all three can produce off-manifold decoder inputs.

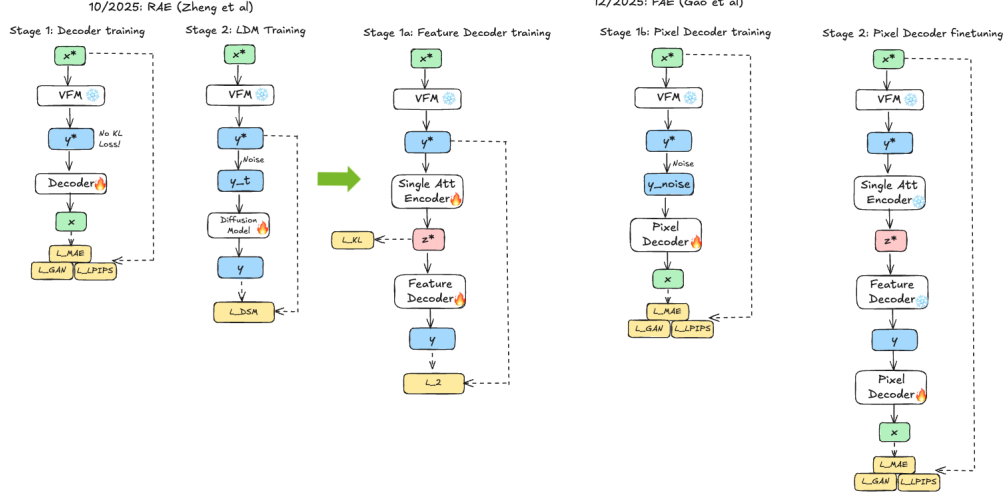


Figure 8: FAE compresses frozen VFM features through a shallow trainable adapter, reconstructs the original feature field with a deeper feature decoder, and separately trains a pixel decoder before fitting the latent prior [Gao et al., 2026b]. The diagram makes clear why “one layer” describes the feature adapter, not the whole system. Author-created Excalidraw diagram.

6 Jointly Learning the Latent Interface

6.1 Unified Latents: rate as a design variable

UNIFIED LATENTS uses a deterministic encoder followed by fixed Gaussian noise, a diffusion prior whose minimum noise matches the encoder noise, and a diffusion decoder [Heek et al., 2026]. The prior is trained with an unweighted diffusion ELBO, while reweighting the decoder ELBO controls which information remains in the latent. This construction makes rate explicit rather than treating latent dimension as its proxy.

For any auxiliary code distribution $q_\phi(z | x)$ and prior $p_\theta(z)$,

$$\mathbb{E}_{p(x)} D_{\text{KL}}(q_\phi(z | x) \| p_\theta(z)) = \underbrace{I_q(X; Z)}_{\text{information rate}} + \underbrace{D_{\text{KL}}(q_\phi(z) \| p_\theta(z))}_{\text{aggregate mismatch}}, \quad (7)$$

where $q_\phi(z) = \int p(x) q_\phi(z | x) dx$. The expected prior cost upper-bounds mutual information because the second term is non-negative. It simultaneously charges for information in each code and for a prior that fails to match the aggregate posterior. UL operationalizes this relation with diffusion-based likelihood terms.

The nuance matters: encoder, prior, and decoder are co-trained in the first stage, but UL’s strongest reported generation configuration freezes the learned autoencoder and trains a fresh prior afterward. This is evidence that joint training can learn a useful interface, not evidence that a single training job is always optimal.

6.2 UNITE: shared weights with routed gradients

UNITE uses one Generative Encoder in two modes: tokenization of clean data and denoising of corrupted latent tokens [Duggal et al., 2026]. Register tokens create a compact code, and the same core weights participate in representation formation and latent denoising. Yet the best configuration stops the denoising gradient at the clean latent. That stop-gradient does not make the tasks independent: the denoising mode still updates shared parameters, which subsequently changes tokenization. The coupling is *through shared weights across modes*, not through direct differentiation of the prior loss through the clean-code computation.

This distinction explains why two statements can both be true: UNITE is an end-to-end shared model, and naive direct latent gradients can be harmful. The architecture couples the tasks while the

Method	Representation source	Prior gradient to clean encoder	Anti-collapse / control	Final prior protocol	Most precise description
REPA-E	VAE encoder aligned to frozen VFM	✗	alignment and re-construction	jointly scheduled DiT	encoder adaptation with routed gradients
UL	learned noisy continuous code	✓	fixed noise and variational rate	strongest result retrain prior	joint interface learning with explicit rate
UNITE	shared Generative Encoder/registers	✗ (best setting)	stop-gradient and mode sharing	shared latent denoiser	shared parameters, selectively routed gradient
GenFirst	learned continuous code	✓	entropy term and curriculum	fresh prior also evaluated	prior-shaped representation with collapse control

Table 2: Why end-to-end is not a sufficient method description.

stop-gradient controls a high-variance or collapse-prone route. UNITE also reports experiments on images, QM9 molecules, and MP20 crystals, making it unusually relevant to cross-domain transfer.

6.3 GenFirst: give the prior a gradient, then prevent collapse

GENFIRST explicitly allows the generative prior to shape the encoder [Zheng et al., 2026]. Its diagnosis is that naive joint likelihood training contains pressure to reduce the entropy of the aggregate code because a concentrated distribution is easier for the prior. A schematic objective is

$$\mathcal{J} = \underbrace{\lambda_{\text{rec}} \mathcal{L}_{\text{rec}}}_{\text{retain information}} + \underbrace{\lambda_{\text{fit}} \mathcal{L}_{\text{prior}}}_{\text{fit aggregate codes}} - \underbrace{\lambda_H H(q_\phi(z))}_{\text{oppose collapse}}, \quad (8)$$

combined with a generation-first schedule: prioritize a modelable latent before tightening reconstruction. The entropy term is not decorative. Without a countervailing force, the prior objective can be improved by mapping diverse x to similar z .

GenFirst and UL therefore share the ambition to make rate and generation pressure part of representation learning, but their mechanisms differ. UL derives a variational rate-control construction with fixed encoder noise; GenFirst foregrounds entropy balance and curriculum. GenFirst, too, evaluates a freshly trained prior after learning the autoencoder, so end-to-end latent discovery and final prior training should be reported separately.

7 What Makes a Representation Generative?

We propose reporting six partly independent properties.

1. Reconstruction fidelity. Measure distortion in pixels and perceptual features, but also editing consistency and fine-structure recovery. Clean reconstruction tests $D(E(x))$, not $D(\tilde{z})$ for prior samples $\tilde{z}$.

2. Semantic organization. Linear probes, retrieval, segmentation, and robustness reveal whether downstream factors are accessible. They do not establish invertibility. Moreover, spatial and global semantics play different roles: patchwise denoiser alignment rewards spatial correspondence, while a latent prior may benefit from globally meaningful organization.

3. Information rate. Latent dimension alone is not rate. A continuous $64 \times 64 \times 1024$ tensor can carry little information if heavily noised, or enormous information at high precision. Report an entropy or coding proxy, encoder noise, quantization, and the prior cross-entropy where possible. Equation (7) provides a principled decomposition.

4. Distributional modelability. Whitening, channel scale, token outliers, covariance condition number, and tail behavior affect optimization. A generator can fail on a perfectly reconstructive latent

Property	Useful measurements	Common proxy failure	Representative interventions
Fidelity	PSNR/SSIM, rFID, LPIPS, edit recovery	clean recon ignores sampled latents	residual detail paths, richer decoder
Semantics	probes, retrieval, dense tasks	global probe hides spatial mismatch	REPA, multi-layer VFM features
Rate	nats/bits, prior NLL, encoder noise	tensor size is not information	UL weighting/noise, quantization
Modelability	prior loss, covariance, tails, convergence	rFID can improve while gFID worsens	normalization, compact bottleneck
Geometry / spectrum	norm/radius, geodesic tests, PSD	Euclidean metrics ignore support	spherical codes, spectrum matching
Robustness	drifted-latent recon, Jacobian gain	$D(E(x))$ is in-distribution only	noisy-feature or sampled-code training

Table 3: A minimum diagnostic panel for generative representations.

simply because the aggregate distribution is ill-conditioned. Conversely, a low-dimensional smooth code may be easy to model while discarding details the decoder cannot recover.

5. Geometry and spectrum. If features concentrate on a sphere or another manifold, Euclidean interpolation can traverse low-density interiors. Hyperspherical methods such as SPHERE ENCODER and DRAE make the geometry explicit [Yue et al., 2026, Ma et al., 2026]. Frequency organization is another view. SPECTRUM MATCHING proposes that a diffusion-friendly encoder should expose a suitably flattened power spectrum and that decoder training should pair image- and latent-frequency perturbations [Ning et al., 2026].

The exact foundation is Parseval’s identity. For an orthonormal transform F ,

$$\begin{aligned}
\|x - \hat{x}\|_2^2 &= \|F(x - \hat{x})\|_2^2 \\
&= \sum_k \underbrace{|(Fx)_k - (F\hat{x})_k|^2}_{\text{error at frequency } k}.
\end{aligned} \tag{9}$$

Thus pixel MSE decomposes exactly by frequency. It does *not* follow as a theorem that a particular flattened spectrum is sufficient or optimal for diffusion. The stronger spectrum-matching hypothesis is an empirically supported design hypothesis. It is complementary to rate and manifold geometry, not a replacement for them.

6. Decoder robustness. At inference, the decoder sees samples from the learned prior, not exact encoder outputs. Let $\delta = \tilde{z} - E(x)$. Locally,

$$D(E(x) + \delta) \approx D(E(x)) + J_D(E(x))\delta.$$

Even small latent error becomes large pixel error when the decoder Jacobian has high gain along likely prior-error directions. Noise injection, drifted-latent training, consistency losses, and explicit Jacobian control all target this mismatch. The right perturbation distribution should resemble actual prior errors rather than isotropic noise by default.

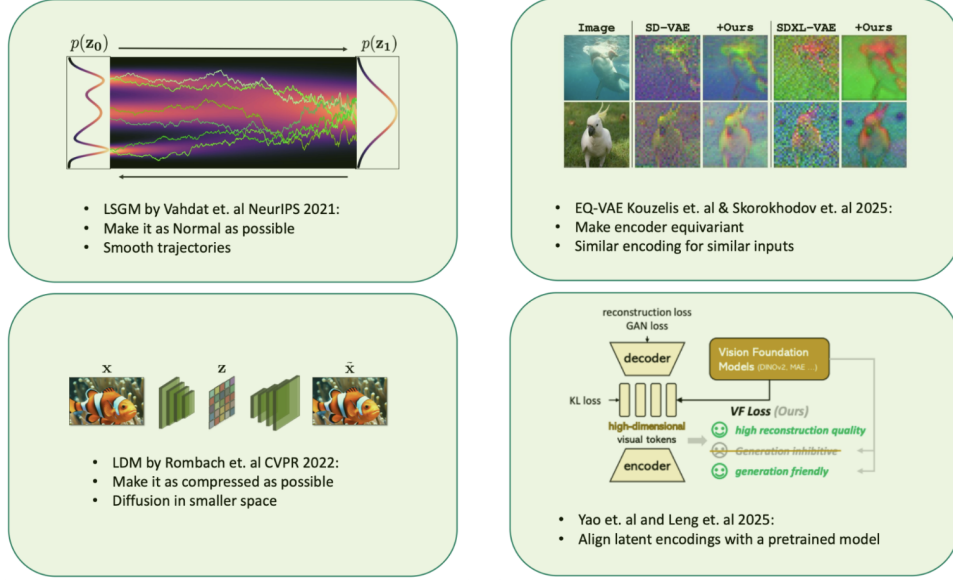


Figure 9: Earlier intuitions about a “good latent space” expressed as four design motifs: smooth prior geometry, equivariant or perturbation-stable codes, computational compression, and semantic alignment. The six-property framework in Table 3 extends this picture by separating rate, modelability, geometry, and decoder robustness. Author-created Excalidraw composition; small panels adapted from the cited source works remain copyright of their original authors and publishers and are excluded from this paper’s CC BY license.

8 Native and Self-Supervised Generative Representations

8.1 Sphere Encoder and iterative refinement

SPHERE ENCODER learns a conditionally uniform hyperspherical latent and samples directly from the sphere [Yue et al., 2026]. It can generate in one decoder pass or refine by repeatedly encoding and decoding. This is not an RAE: the code is not inherited from a pretrained semantic encoder. It is also not a diffusion prior: the intent is to make a simple uniform prior directly usable. Its recurrence is best seen as learned projection or refinement around an autoencoder.

8.2 Latent Forcing: order the information reveal

LATENT FORCING keeps pixel-space generation but jointly diffuses pixels and pretrained semantic features with separate time schedules [Baade et al., 2026]. Denote noisy pixels x_{t_x} and noisy semantic features r_{t_r} . A joint network predicts both flows:

$$\mathcal{L} = \mathbb{E}[\|v_x(x_{t_x}, r_{t_r}, t_x, t_r) - u_x\|^2 + \lambda_r \|v_r(x_{t_x}, r_{t_r}, t_x, t_r) - u_r\|^2].$$

By choosing t_r to resolve semantics before t_x resolves pixels, the trajectory exposes a coarse semantic plan before high-frequency synthesis. This is trajectory design, not tokenizer learning. The paper’s strongest schedule should not be generalized into a universal rule without considering modality and architecture.

8.3 Self-Flow: the teacher is generated internally

SELF-FLOW removes the external semantic teacher by creating asymmetric views of the same noisy sample [Chefer et al., 2026]. Tokens receive heterogeneous noise times. An exponential-moving-average (EMA) teacher observes a cleaner version, while the student observes mixed noise and

predicts teacher features in addition to the flow target. Schematically,

$$h^- = f_{\bar{\theta}}(x_{\min(t,s)}, \min(t, s)), \quad (10)$$

$$\mathcal{L}_{\text{SF}} = \underbrace{\|v_{\theta}(x_t, t) - u_t\|^2}_{\text{flow}} + \lambda \underbrace{[1 - \cos(g(h_{\theta}(x_t, t, s)), \text{sg}(h^-))]}_{\text{internal feature target}}. \quad (11)$$

The EMA, stop-gradient, and cleaner-context asymmetry prevent the target from trivially chasing the student. The method spans image, video, audio, multimodal, and action experiments, with modality-specific schedules. It removes dependence on an external VFM; it does not imply the absence of an autoencoder or tokenizer in every implementation.

Product-scale evidence. Black Forest Labs states that FLUX.3 builds on Self-Flow and uses a shared flow-matching backbone across image, video, audio, and action generation [Black Forest Labs, 2026]. This is relevant evidence that the design family is being scaled, but the public report describes evaluations as preliminary and does not disclose enough detail for independent reproduction. It should be cited as a first-party system announcement, not counted as a controlled research comparison.

9 Beyond Images

9.1 Language and unified multimodal models

LDLM jointly adapts pretrained language features, a latent diffusion model, and a decoder [Meshchaninov et al., 2026]. It uses a hidden-state reconstruction loss, a warm-up that delays difficult coupling, adaptive timestep sampling, and noise on decoder inputs. These stabilizers mirror the visual literature: control the gradient path, teach the decoder about inference-time drift, and allocate optimization effort where the prior is weakest.

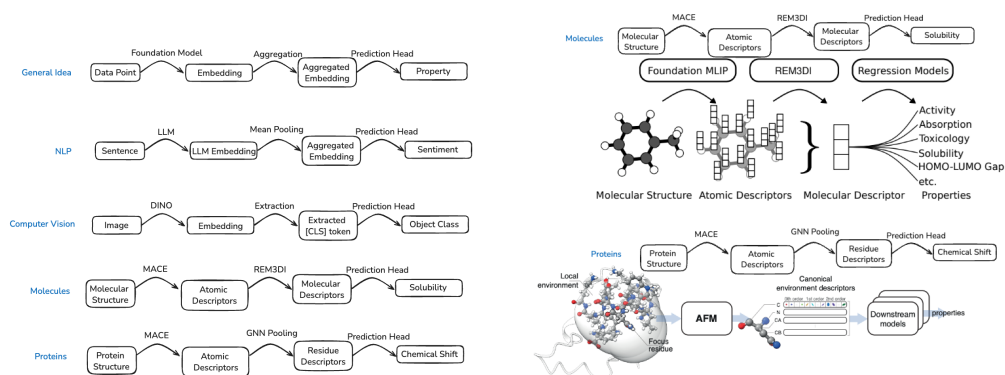
REPR-ALIGN takes the opposite direction, adapting an autoregressive language model toward diffusion through representation alignment rather than fully retraining it [Peng et al., 2026a]. UNIR argues that a shared context-visual tokenizer is central to unified autoregressive understanding and generation [Peng et al., 2026b]. Controlled 2026 studies report that generation and understanding can transfer when they require shared knowledge, but also identify asymmetric interference when both are forced through identical computation paths [Zhu et al., 2026, Wu et al., 2026]. Unified should therefore be decomposed into shared vocabulary, shared latent, shared trunk, shared objective, and shared inference procedure.

9.2 Molecules, proteins, and 3D scenes

Structural domains sharpen the interface problem because validity is geometric. Rotations and translations should not change identity; atom order may be arbitrary; chirality matters; structures vary in size; and a small local coordinate error can propagate globally. A semantic image feature can discard texture harmlessly, whereas a protein code that discards stereochemistry is chemically wrong.

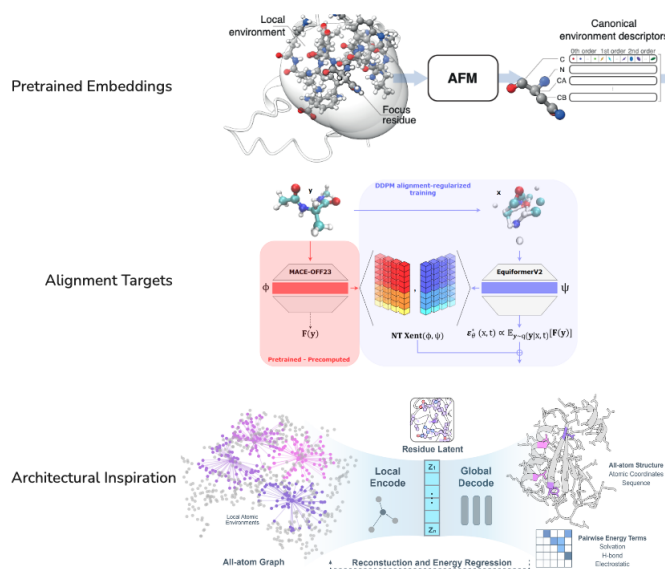
UNITE reports the same shared tokenization/denoising mechanism on QM9 molecules and MP20 crystals as well as images [Duggal et al., 2026]. KANZI uses a flow-matching decoder as a compact protein-structure tokenizer and reports strong reconstruction with a much smaller model than specialized geometric tokenizers [Dilip et al., 2026]. ONESWORLD builds a 3D unified representation autoencoder from pretrained 3D features, adds cross-view consistency, and trains on both clean and drifted representations [Gao et al., 2026a]. These examples support the transfer of *principles*—rate, geometry, and decoder robustness—not the literal reuse of image architectures.

Atomistic foundation models provide a second bridge. MACE embeddings transfer from energy/force pretraining to molecular and protein property prediction [Batatia et al., 2025, Wedig et al., 2025, Bojan et al., 2025]. As in language and vision, a local feature field must be aggregated into a task-level representation. The generative use cases are more tentative: representation alignment has been explored for force prediction and conformation generation [Pinede et al., 2025], while direct conditioning on atomistic embeddings has not uniformly improved structure prediction [Corley et al., 2025]. This distinction between transferable features, aligned generators, and architectural inspiration prevents a downstream probe result from being overread as a generative result.



(a) Local MACE descriptors are aggregated into molecular representations for property prediction. (b) The same local-to-global pattern across NLP, vision, molecules, and proteins.

How to use atomistic foundation models?



(c) Three roles for atomistic foundation models: pretrained embeddings, alignment targets, and architectural inspiration.

Figure 10: How the visual representation-generation taxonomy transfers to atomistic systems [Batatia et al., 2025, Wedig et al., 2025, Bojan et al., 2025, Pinede et al., 2025]. Author-created Excalidraw compositions. Embedded molecular and protein renderings or small source-paper panels remain copyright of their original authors and publishers and are excluded from this paper's CC BY license.

Protein modeling also clarifies the difference between representation recurrence and generative self-conditioning. AlphaFold recycling repeatedly feeds predicted structure-derived features back through the network to refine one prediction [Jumper et al., 2021]; ESMFold uses a related folding trunk around language-model features [Lin et al., 2023]. RFdiffusion uses diffusion self-conditioning in structure generation [Watson et al., 2023]. The generic diffusion technique predates RFdiffusion: Analog Bits explicitly introduced self-conditioning of a model’s previous clean-data estimate for discrete diffusion [Chen et al., 2023]. Recycling, self-conditioning, and iterative decoding all feed predictions forward, but differ in state, objective, and stochastic semantics. Appendix B makes the distinction precise.

10 Evaluation and Reporting

10.1 Why headline FID is insufficient

Generation FID depends on training data, resolution, sample count, guidance scale, number of function evaluations, sampler, model size, augmentation, and evaluation implementation. Reconstruction FID depends on the decoder and on whether stochasticity is used. Reporting a lower number across papers does not identify which representation is better unless these variables are controlled.

At minimum, a representation paper should report:

1. clean reconstruction and reconstruction under realistic latent drift;
2. unconditional or class/text-conditional generation with and without guidance;
3. latent rate or a defensible proxy, spatial size, channel count, normalization, and encoder noise;
4. prior convergence curves at matched compute, not only the best checkpoint;
5. semantic and dense/spatial probes where relevant;
6. all stop-gradients, frozen modules, EMA targets, warm-ups, and stage boundaries; and
7. final-prior protocol, especially whether a new prior is trained after joint latent learning.

10.2 Ablations that answer causal questions

An informative ablation changes one interface property while holding capacity and compute fixed. Examples include whitening versus raw features; identical decoder with clean versus drifted-latent training; equal-rate latents with different dimensions; direct prior-to-encoder gradient versus stop-gradient under the same shared trunk; and aligned targets with preserved versus shuffled spatial structure. Scaling only the denoiser may hide a difficult latent rather than repair it.

The most useful negative results are boundary conditions. If representation alignment stops helping after early training, that locates its role as curriculum. If a VFM latent helps a wide DiT but not a narrow one, architecture and representation are coupled. If fresh-prior retraining improves a jointly learned system, that distinguishes interface discovery from final density estimation. Such findings are more reusable than a single champion configuration.

11 Synthesis: What the 2026 Wave Changes

The 2026 papers do not replace the earlier story; they split several overloaded concepts.

End-to-end is a family of gradient graphs. REPA-E aligns the encoder but blocks the diffusion gradient. UNITE shares weights across tokenization and denoising but performs best with a stopped clean code. UL co-trains encoder, prior, and decoder under a variational rate construction, then obtains its strongest generation with a fresh prior. GenFirst deliberately transmits prior pressure to the encoder and introduces entropy and curriculum to prevent collapse. These are not cosmetic variants.

Representation quality is multidimensional. SPECTRUM MATCHING adds frequency organization; spherical methods add explicit topology; UL adds rate; LV-RAE and OneWorld add decoder robustness; RAE-AR adds sequential exposure bias. No one diagnostic subsumes the others.

Internal teachers reduce dependence on external encoders. Self-Flow shows that heterogeneous-noise prediction and an EMA teacher can create useful targets within the generator. Latent Forcing shows that a pretrained semantic track can instead order the generation trajectory. Their shared lesson is controlled information asymmetry, not that all representations should be discarded.

Unification benefits from specialization with communication. Shared representations can transfer knowledge between understanding and generation, but fully identical paths can interfere. A common interface with mode-specific heads, schedules, or routed gradients may be more effective than one undifferentiated state.

12 Limitations and Open Problems

This survey reflects a rapidly moving literature and a September 2026 cutoff. Many works have not completed peer review; code availability and reported status may change. We emphasize primary papers and official project material, but cannot normalize every training and evaluation protocol. Our taxonomy assigns a dominant mechanism, which necessarily compresses hybrids.

Several research questions appear durable:

1. Can latent rate, prior difficulty, and decoder robustness be measured in a common evaluation rather than inferred from architecture?
2. Which prior-error distribution should a decoder train against, and can it be estimated online without destabilizing joint learning?
3. When does a curved manifold require a geometry-aware probability path rather than normalization or increased model width?
4. Can a joint objective adapt semantic features while certifiably preserving downstream capabilities?
5. What is the correct information ordering for video, audio, language, and 3D structures?
6. How should unified systems allocate shared versus specialized capacity as modalities and tasks scale?

13 Conclusion

The field is converging on a simple but demanding idea: a generative latent is an interface, not a by-product. Alignment methods change internal optimization; tokenizer methods decide which information is retained; RAEs inherit semantic geometry and its defects; joint methods let the prior reshape the code but must control collapse; native self-supervision creates targets from the generative process itself.

The practical rule is to inspect three objects together: the information carried by z , the errors made by the prior, and the decoder’s response to those errors. Then trace every gradient. This view explains why reconstruction, semantics, spectrum, geometry, and rate can each matter without any one becoming the universal answer. It also scales beyond images: the constraints change, but the interface-design problem remains.

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A Detailed Derivations

A.1 The VAE bound, line by line

Starting from the marginal likelihood and multiplying by one,

$$\log p(x) = \log \int p(x, z) dz \quad (\text{marginalize } z) \quad (12)$$

$$= \log \int q_\phi(z | x) \frac{p(x, z)}{q_\phi(z | x)} dz \quad (\text{importance identity}) \quad (13)$$

$$= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\psi(x | z)p(z)}{q_\phi(z | x)} \right] \quad (14)$$

$$\geq \mathbb{E}_{q_\phi(z|x)} [\log p_\psi(x | z) + \log p(z) - \log q_\phi(z | x)] \quad (\text{Jensen}) \quad (15)$$

$$= \underbrace{\mathbb{E}_q[\log p_\psi(x | z)]}_{\text{decoder likelihood}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) \| p(z))}_{\text{rate and regularity}}. \quad (16)$$

The gap equals $D_{\text{KL}}(q_\phi(z | x) \| p(z | x))$, so the bound is tight when the approximate posterior matches the true posterior. Intuitively, the encoder sends a message z ; reconstruction rewards useful bits, while the KL charges for messages that are surprising under the prior.

A.2 Why the expected prior cost contains mutual information

Let $q(x, z) = p_{\text{data}}(x)q_\phi(z | x)$. Add and subtract $\log q_\phi(z)$:

$$\mathbb{E}_{p(x)} D_{\text{KL}}(q(z | x) \| p(z)) = \mathbb{E}_{q(x, z)} \left[\log \frac{q(z | x)}{p(z)} \right] \quad (17)$$

$$= \mathbb{E}_{q(x, z)} \left[\underbrace{\log \frac{q(z | x)}{q(z)}}_{\text{mutual information}} + \underbrace{\log \frac{q(z)}{p(z)}}_{\text{aggregate mismatch}} \right] \quad (18)$$

$$= I_q(X; Z) + D_{\text{KL}}(q(z) \| p(z)). \quad (19)$$

The second term vanishes only when the prior exactly matches the aggregate posterior. A stronger learned prior can reduce aggregate mismatch without reducing the information rate. This is why latent dimension and prior loss should not be used interchangeably.

A.3 Noise prediction, clean prediction, and velocity

From $z_t = \alpha_t z_0 + \sigma_t \epsilon$, any two of z_t, z_0, ϵ determine the third:

$$\hat{z}_0 = \frac{z_t - \sigma_t \hat{\epsilon}}{\alpha_t}, \quad \hat{\epsilon} = \frac{z_t - \alpha_t \hat{z}_0}{\sigma_t}.$$

Define velocity $v_t = \alpha_t \epsilon - \sigma_t z_0$. Then

$$\begin{bmatrix} z_t \\ v_t \end{bmatrix} = \begin{bmatrix} \alpha_t & \sigma_t \\ -\sigma_t & \alpha_t \end{bmatrix} \begin{bmatrix} z_0 \\ \epsilon \end{bmatrix}, \quad \begin{bmatrix} z_0 \\ \epsilon \end{bmatrix} = \begin{bmatrix} \alpha_t & -\sigma_t \\ \sigma_t & \alpha_t \end{bmatrix} \begin{bmatrix} z_t \\ v_t \end{bmatrix}.$$

The matrix is orthogonal when $\alpha_t^2 + \sigma_t^2 = 1$. Thus the parameterizations encode equivalent targets at population optimum, but their finite-capacity weighting and numerical conditioning differ by timestep.

A.4 Why naive prior pressure can collapse a learned code

Consider a deterministic encoder $z = E_\phi(x)$ and a flexible prior trained by negative log likelihood,

$$\mathcal{L}_{\text{prior}} = -\mathbb{E}_x \log p_\theta(E_\phi(x)).$$

If the decoder term is weak, mapping every input to a constant z_* makes the aggregate code maximally concentrated and easy for the prior to fit. A reconstruction loss resists because one code cannot reconstruct all inputs. An entropy reward resists directly:

$$\mathcal{J} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{fit}} \mathcal{L}_{\text{prior}} - \lambda_H H(q_\phi(z)).$$

This toy argument does not prove a particular algorithm stable; it explains why letting the prior shape the encoder requires an opposing information-preservation mechanism.

A.5 Spectrum matching and the limit of Parseval

Let F be an orthonormal discrete Fourier or cosine transform, so $F^\top F = I$. Then

$$\|x - \hat{x}\|_2^2 = (x - \hat{x})^\top (x - \hat{x}) \tag{20}$$

$$= (x - \hat{x})^\top F^\top F (x - \hat{x}) \tag{orthonormality} \tag{21}$$

$$= \|F(x - \hat{x})\|_2^2 = \sum_k |(Fx)_k - (F\hat{x})_k|^2. \tag{22}$$

If a binary frequency mask M is applied consistently, the loss becomes

$$\|M \odot F(x - \hat{x})\|_2^2 = \sum_{k: M_k=1} |(Fx)_k - (F\hat{x})_k|^2.$$

This exactly justifies frequency-resolved reconstruction. It does not identify the optimal latent power spectral density. That step requires assumptions about the data, model class, noise schedule, and optimization, or empirical validation.

A.6 Decoder robustness as a directional problem

For prior sample error δ , first-order propagation gives

$$\mathbb{E}\|D(z + \delta) - D(z)\|_2^2 \approx \mathbb{E}[\delta^\top J_D(z)^\top J_D(z) \delta].$$

If $\Sigma_\delta = \mathbb{E}[\delta \delta^\top]$, then

$$\mathbb{E}\|J_D \delta\|^2 = \text{tr}(J_D^\top J_D \Sigma_\delta).$$

Isotropic noise training uses $\Sigma_\delta = \sigma^2 I$ and penalizes average Jacobian energy. Actual prior errors can be anisotropic, concentrating along difficult directions. Sampling the current prior or fitting a local error covariance can therefore train the decoder more efficiently than indiscriminate isotropic perturbation.

B Self-Conditioning, Recycling, Looping, and Recurrence

These labels describe a shared computational motif—feed an intermediate prediction back into later computation—but not a shared probabilistic object.

Self-conditioning in diffusion. At step t , a denoiser consumes a previous estimate of the clean sample,

$$\hat{x}_0^{(t)} = f_\theta(x_t, t, \text{sg}(\hat{x}_0^{(t+1)})).$$

Analog Bits used this technique for discrete-data diffusion before RFDiffusion adopted it for protein structures [Chen et al., 2023, Watson et al., 2023]. The stop-gradient means the previous estimate acts as an auxiliary condition during ordinary training. It changes the parameterized reverse kernel while preserving a valid generative procedure if the augmented state and training/inference conditioning are defined consistently. What is usually lost is the simplicity of a Markov kernel written only as $p_\theta(x_{t-1} \mid x_t)$: the implementation depends on cached model state. It does not automatically invalidate diffusion, but likelihood or ELBO statements derived for the unaugmented Markov chain cannot be quoted without modification.

Recycling. AlphaFold recycling maps a current structure and pair/single representations back into the folding trunk to refine the same prediction [Jumper et al., 2021]. It is supervised iterative inference, not a stochastic diffusion-time transition. Parameter sharing makes the unfolded computation recurrent. Training may unroll a limited number of recycles and stop gradients through earlier iterations for memory or stability.

Looping and generic recurrence. A looped Transformer reapplies a block $h_{k+1} = F_\theta(h_k, x)$, often with shared weights. Backpropagation through time (BPTT) differentiates through the unrolled chain:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_{k=0}^{K-1} \frac{\partial \mathcal{L}}{\partial h_K} \left(\prod_{j=k+1}^{K-1} \frac{\partial h_{j+1}}{\partial h_j} \right) \frac{\partial h_{k+1}}{\partial \theta}.$$

The Jacobian product explains exploding or vanishing gradients and motivated gated recurrent networks [Hochreiter and Schmidhuber, 1997]. Truncated BPTT stops some paths and optimizes a surrogate.

Deep equilibrium models. A deep equilibrium model defines $h^* = F_\theta(h^*, x)$ and solves for the fixed point rather than storing all iterations [Bai et al., 2019]. Implicit differentiation gives

$$\frac{dh^*}{d\theta} = \left(I - \frac{\partial F_\theta}{\partial h} \Big|_{h^*} \right)^{-1} \frac{\partial F_\theta}{\partial \theta} \Big|_{h^*}.$$

DEQs are therefore a mathematically important ancestor of learned iterative refinement, but most recycling or self-conditioning systems are finite unrollings and do not enforce convergence to a fixed point. The useful connection is shared-weight computation and stability analysis; equality of names is not evidence of an equilibrium model.

Mechanism	Feedback state	Index being iterated	Typical training gradient	Probabilistic role
Diffusion self-conditioning	previous $\hat{x}_0$	noise/sampling time	often stopped through cached estimate	augments reverse-kernel context
Protein recycling	structure, single/pair features	refinement recycle	finite unroll or stopped earlier recycle	iterative supervised inference
Looped Trans-former DEQ	hidden state or tokens equilibrium hidden state	depth step root-solver iteration	BPTT or truncation implicit differentiation	architecture dependent implicit layer, not inherently generative
Sphere Encoder refinement	re-encoded decoded sample	refinement loop	method-specific finite loop	moves sample toward learned data manifold

Table 4: Related computation patterns with different states, objectives, and theoretical commitments.

C Method Cards for the 2026 Additions

Unified Latents [Heek et al., 2026]. Intervention: learned continuous code. Coupling: joint encoder/prior/decoder first stage. Distinctive control: fixed encoder noise matched to the prior’s minimum noise and a reweighted decoder ELBO. Caveat: strongest generation includes fresh-prior retraining.

UNITE [Duggal et al., 2026]. Intervention: one register-token Generative Encoder serves tokenizer and latent-denoising modes. Coupling: shared parameters, with stopped clean-code gradient in the best setting. Evidence: images plus QM9 and MP20. Caveat: shared weights should not be described as unrestricted gradient sharing.

GenFirst [Zheng et al., 2026]. Intervention: jointly learned latent shaped directly by the prior. Coupling: active prior-to-encoder gradient. Distinctive control: entropy counterterm and generation-before-reconstruction schedule. Caveat: code was pending at the survey cutoff, and a fresh final prior remains part of the evaluation.

FAE [Gao et al., 2026b]. Intervention: compact bottleneck on frozen VFM features. Coupling: staged. Distinctive control: one-layer encoder, deeper feature decoder, and separate pixel decoder exposed to noisy features. Caveat: one layer describes the adapter, not the entire autoencoding system.

Spectrum Matching [Ning et al., 2026]. Intervention: encoder latent spectrum and decoder frequency training. Coupling: tokenizer-level. Distinctive control: encoding spectrum matching, decoding spectrum matching, and a Difference-of-Gaussians alignment target. Caveat: the algebraic decomposition is exact; the proposed preferred spectrum is empirical.

Sphere Encoder [Yue et al., 2026]. Intervention: learned hyperspherical code with a simple uniform prior. Coupling: learned generative autoencoder. Distinctive control: conditional uniformity and optional iterative refinement. Caveat: neither a pretrained-feature RAE nor a diffusion prior.

Latent Forcing [Baade et al., 2026]. Intervention: an auxiliary semantic trajectory in pixel diffusion. Coupling: joint dual-track denoising with separately scheduled times. Distinctive control: semantics resolve before pixels. Caveat: no learned tokenizer is implied.

Self-Flow and FLUX.3 [Chefer et al., 2026, Black Forest Labs, 2026]. Intervention: internal EMA representation target from a cleaner view. Coupling: native self-supervised flow training. Distinctive control: heterogeneous token timesteps and cross-time prediction. Caveat: the academic method has technical detail; FLUX.3 is a first-party deployment report with preliminary evaluation.

D Expanded Molecular and Structural Case Study

The six-property framework changes meaning in structural domains:

Fidelity	Bond lengths, local frames, clashes, chirality, and long-range topology matter; a perceptually plausible structure may still be chemically invalid.
Semantics	Functional family, fold, secondary structure, or energy landscape can be more relevant than image-style class labels.
Rate	Variable-size structures complicate per-sample comparisons. Report bits or nats per atom/residue as well as token count.
Modelability	Equivariant coordinate distributions and discrete sequence/structure mixtures require different priors.
Geometry	SE(3) symmetry, periodic boundary conditions, and manifold-valued rotations are first-class constraints rather than optional regularizers.
Robustness	Small local-frame or torsion errors can accumulate along a chain; decoder training should reflect the error modes of the structure prior.

This makes the analogy to visual RAEs useful but incomplete. A pretrained 3D encoder can provide semantics, yet its invariances may remove precisely the degrees of freedom required for generation. Cross-view or equivariant consistency can preserve identity while avoiding arbitrary coordinate frames. Drifted-latent decoder training is especially important when validity depends on coordinated global constraints.

UNITE’s cross-domain experiments suggest that shared register-token interfaces can survive a change in data type [Duggal et al., 2026]. Kanzi suggests that flow-based decoding itself can be an efficient tokenizer design [Dilip et al., 2026]. OneWorld demonstrates pretrained 3D features, cross-view consistency, and exposure to drifted representations [Gao et al., 2026a]. The open question is not whether RAE for molecules works in the abstract, but which invariances, rates, and perturbations correspond to the downstream structural generator.

E Reading-Group Guide

A 60-minute discussion can be organized around five questions:

1. **10 min: What is the interface?** Derive Equations (1) and (7); separate distortion, rate, and aggregate prior mismatch.
2. **12 min: Where does representation enter?** Use Figure 4 to contrast denoiser alignment, tokenizer shaping, RAEs, and native systems.
3. **15 min: What does end-to-end mean?** Trace Figure 2 and Table 2; ask which gradient each stabilizer blocks.
4. **13 min: What makes a latent modelable?** Debate whether rate, geometry, spectrum, or decoder robustness explains each result.
5. **10 min: Does the analogy transfer?** Compare image latents with language and molecular structures, then identify broken invariances.

Suggested core reading. Read REPA [Yu et al., 2024] for auxiliary alignment, RAE [Zheng et al., 2025] for pretrained-feature latents, and either UL [Heek et al., 2026] or GenFirst [Zheng et al., 2026] for joint latent learning. Add Spectrum Matching [Ning et al., 2026] for diagnostics and Self-Flow [Chefer et al., 2026] for native self-supervision. UNITE [Duggal et al., 2026] is the best bridge to non-image domains in this set.